



ICML

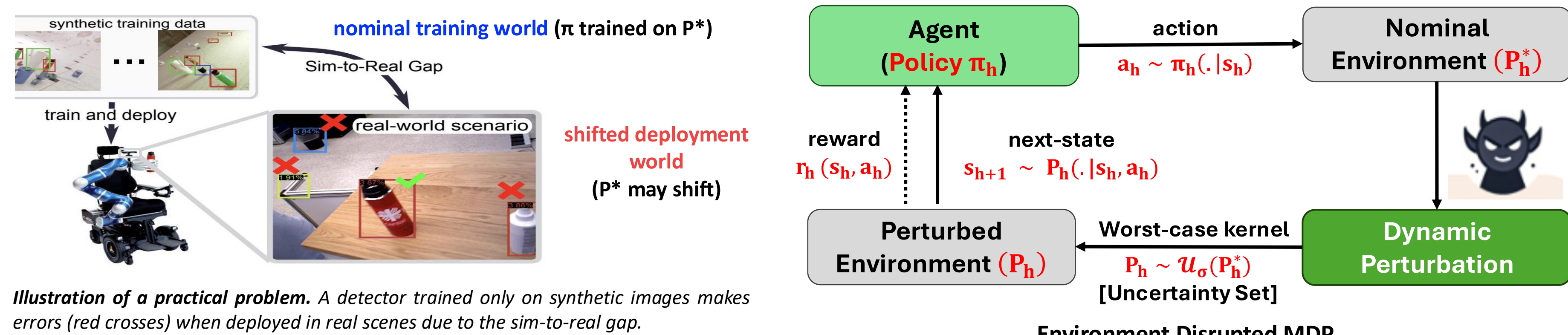
Online Distributionally Robust Reinforcement Learning with General Function Approximation

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Motivation: Bridging the Sim-to-Real Gap



Sim-to-real gap. Real deployments face unpredictable noise, unmodelled perturbations and even adversarial attacks that are not fully captured by the simulator.

Policies that look excellent in simulation can fail catastrophically after deployment.

Vanilla RL

Optimizes expected return under the training dynamics only.

Distributionally Robust RL (DR-RL)

Optimizes performance under a divergence ball around the nominal dynamics.

Online challenge

Data come from P^* , while robust value depends on worst-case dynamics.

Key Challenges: Why is Online Robust RL difficult?

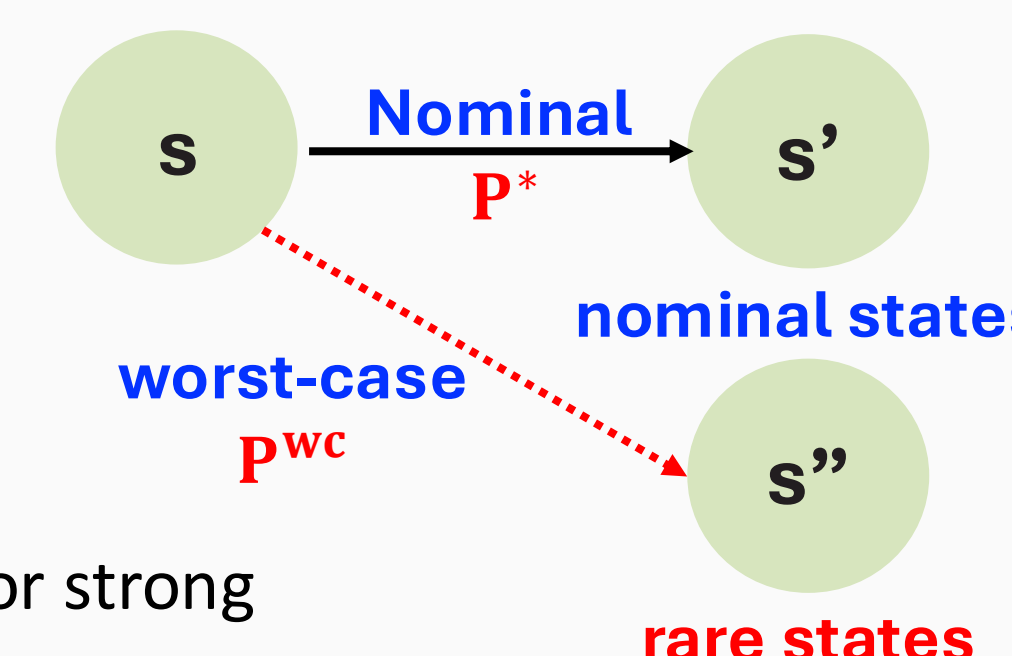
Purely **online** robust reinforcement learning with **general function approximation** is fundamentally hard.

Off-Dynamics Learning

- Data come from **nominal dynamics**
- Robustness is evaluated under **worst-case dynamics**.

Key Challenge: Distribution mismatch Information bottleneck (He et al. 2025)

Existing methods need generative model or rich offline data or strong coverage assumptions (Shi and Chi 2024).



Scalability

Real-world robust RL require: • neural networks • continuous states • large action spaces

Key Challenge:

- Tabular bonuses scale with $|\mathcal{S}| |\mathcal{A}|$.
- Linear function approximation methods require restrictive linearly structure assumptions.

No general intrinsic-complexity theory existed for online DR-RL.

Robust Bellman Learning

Robust Bellman update requires optimization over uncertainty sets: $(\mathcal{T}_h^{\phi, \sigma} \mathbf{v})(s, a) = r(s, a) + \inf_{P \in \mathcal{U}_{\phi, \sigma}} E_P[V]$.

Existing Standard fitted RL/UCB assume nominal Bellman updates (Jin et al., 2021);

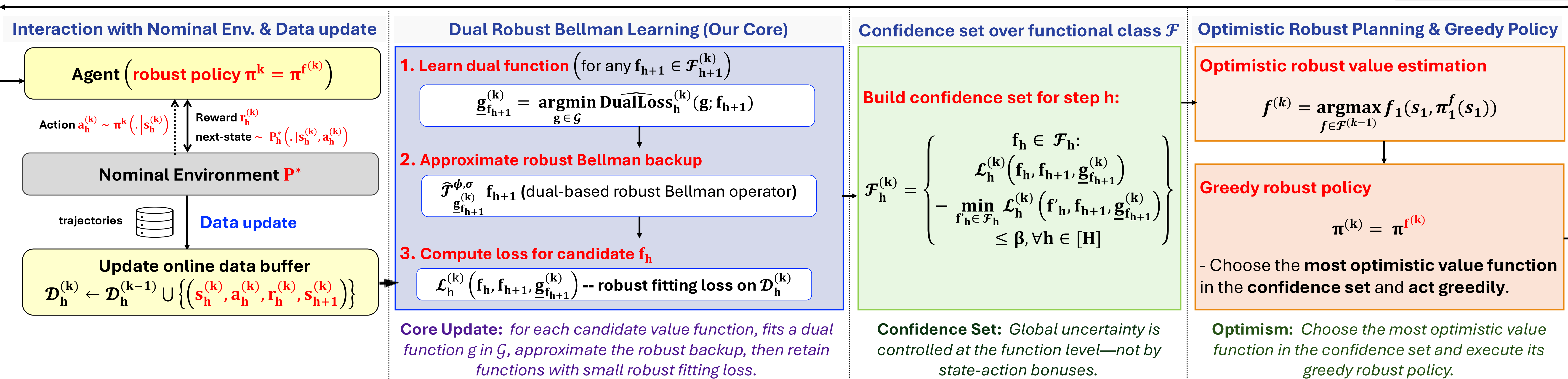
Robust RL requires learning a worst-case Bellman operator online (Panaganti et al., 2022).

- Key Challenge:**
- Worst-case operator is not directly observable
 - Inner optimization required at every (s, a) .

Can we learn a robust policy purely online, with general function approximation, and provable regret?

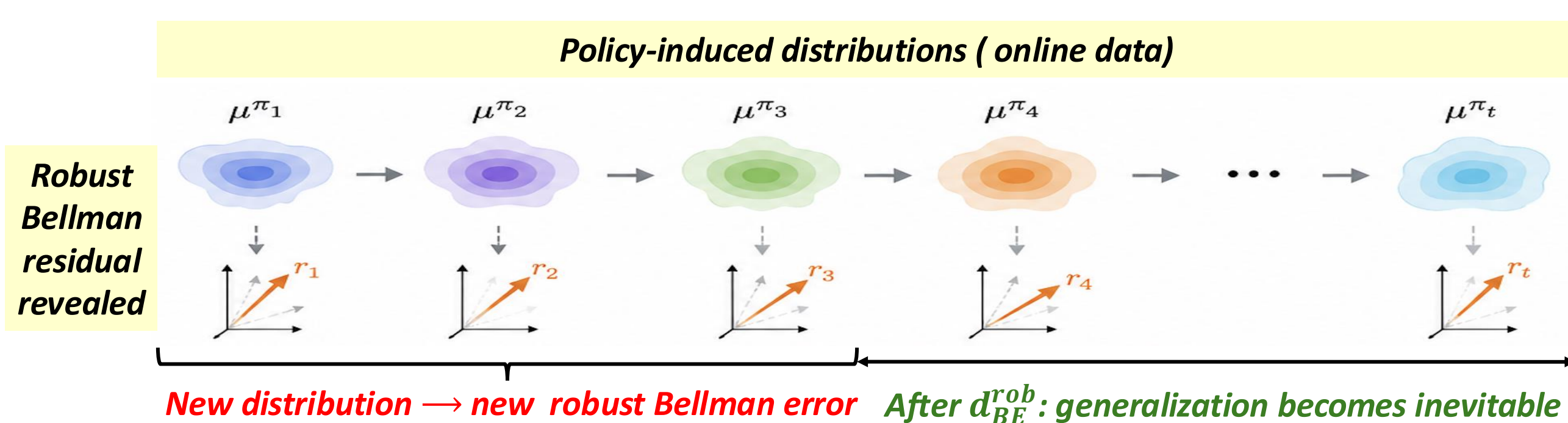
Algorithm: Robust Fitted Learning with ϕ -Divergence (RFL- ϕ)

Input • $\mathcal{F}$: value functional class • $\mathcal{G}$: dual functional class • β : confidence parameter • σ : robustness radius



Theoretical Guarantee: Why this matters?

Robust Bellman-Eluder (BE) dimension: $d_{BE}^{rob} = \dim_{DE}((\mathcal{I} - \mathcal{T}_h^{\phi, \sigma})\mathcal{F}) \Rightarrow$ captures the exploration complexity under robust Bellman updates



Counts how many policy distributions reveal new, statistically, independent Bellman errors.

Why it matters?

- Smaller $d_{BE}^{rob} \Rightarrow$ fewer essentially different robust Bellman errors.
- Exploration scales with d_{BE}^{rob} not $|\mathcal{S}| |\mathcal{A}| \Rightarrow$ sample efficient.
- First regret guarantee for purely online DR-RL with general function approximation.

approximation: $\text{Regret}(K) \leq \tilde{O}\left(\sqrt{d_{BE}^{rob} H^2 B_{\phi}^2(\sigma) K} + \epsilon^{\text{dual}}\right)$.

- Recovers near-optimal tabular and linear robust RL guarantees.

Key Takeaway:

Robust fitted Bellman learning achieves strong robustness under severe distribution shifts without relying on tabular representations.

Acknowledgements

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- He, Y., Liu, Z., Wang, W., and Xu, P. 2025. Sample Complexity of Distributionally Robust Off-Dynamics Reinforcement Learning with Online Interaction. In *Forty-second International Conference on Machine Learning*.
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References

Experiments: Cartpole

Strong robustness under perturbations

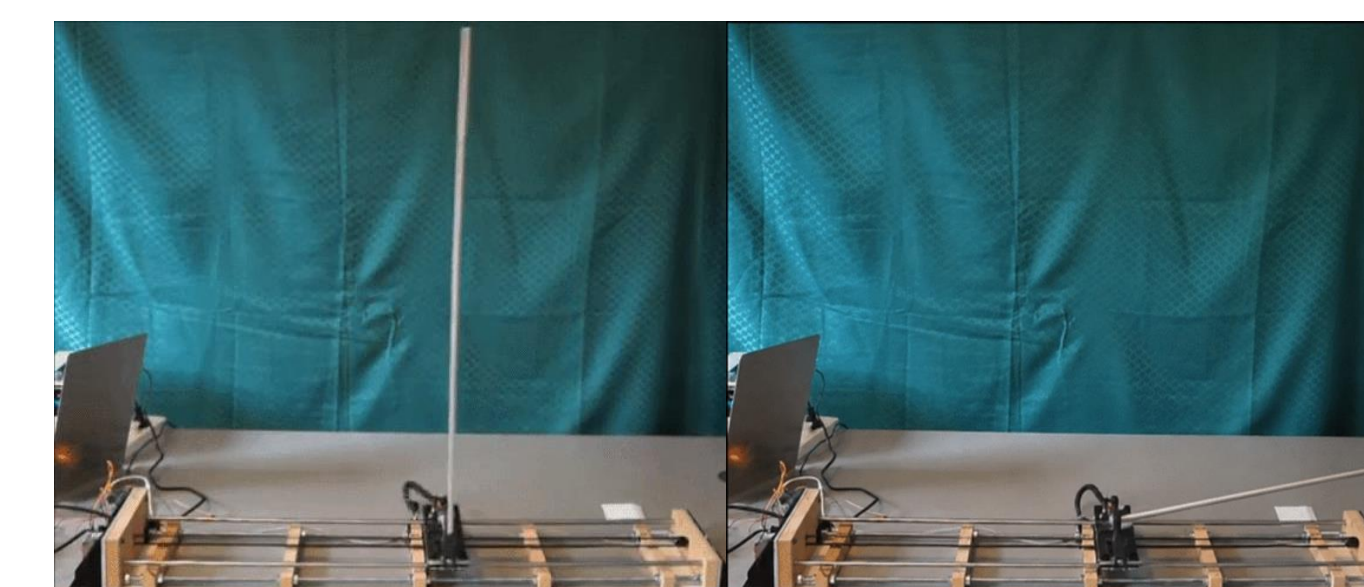
- Up to $2 \times$ higher reward than DQN;
- competitive with tabular OPROVI-TV.

Stable under severe dynamics shifts

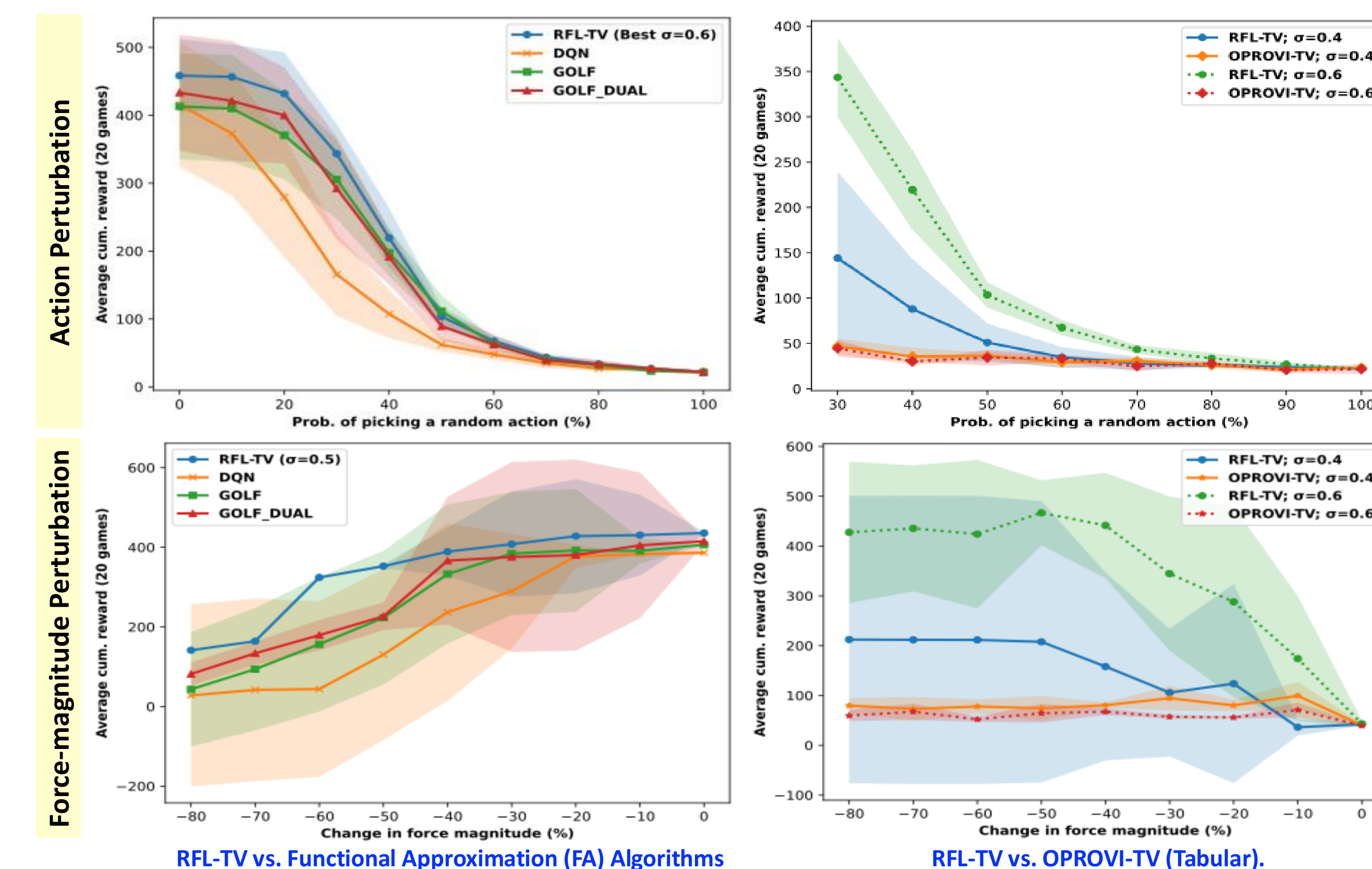
- Robust even under $-80\% \sim -40\%$ force perturbations;
- non-robust FA baselines collapse.

Moderate runtime overhead

- Only $\sim 1.5 \times$ DQN despite robust Bellman learning.



CartPole policies trained nominally, then tested under action and dynamics perturbations.



Poster Q&A & Follow-Up:
<https://forms.gle/FRMRDh4Taj2GY5wC7>